

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use. All numbers: synthetic data.

The six items in one sentence each

1. **Proposed system:** for an adult on metformin, compare SGLT2i, DPP-4i and sulfonylurea by each one's 6-month HbA1c change (plus weight and low-sugar risk) with 95% ranges, after cited safety rules remove unsafe options; explained with cited WHO/FDA passages; the clinician decides.
2. **Hardware/software:** any 8 GB laptop; Python 3.12 + scikit-learn + FastAPI; phone website on Netlify with Supabase accounts; all free.
3. **Timeline:** scope and design (Jul-Sep); chat app, engine, website and RAG done by 26 Sept (ahead of plan); integration, clinician review, embedding search in Oct; results frozen 16 Oct; paper and report by 30 Oct.
4. **Design:** safety rules → propensity/overlap → AIPW/DR-learner → cost → cited evidence → clinician; audit log without patient values.
5. **Implementation:** secure chat app (guards, CAPTCHA+MFA, patient panel, guardrails, voice, Docker; 630 tests); causal engine + benchmark + refutation + secondary outcomes; live website with MFA and approval; RAG with WHO 2018 + 7 FDA notices, quoted/Gemini/Ollama explanations and a citation checker; 201 engine/RAG/web tests.
6. **Results:** naive bias $-0.159 \rightarrow$ AIPW $+0.002$; AIPW coverage 90–95%; per-patient coverage 92.7–94.7%; regret 0.011 vs 0.062; balance 0.70 $\rightarrow$ 0.08; refutation 9/9. RAG (60 questions): recall@5 0.933, citation precision 1.0, 0 dose leaks, abstention 0.80 (target 0.95).

Key words, one line each

Confounding by indication: sicker patients got SGLT2i more, so plain comparisons mislead. **Propensity:** chance a patient like this got each drug (multinomial logistic, cross-fitted). **Overlap/abstain:** propensity $< 0.05 \Rightarrow$ “insufficient evidence”, never a guess. **IPW:** weight each patient by 1/propensity so groups look alike. **AIPW (doubly robust):** outcome-model guess + weighted correction; right if either model is right. **DR-learner:** regress AIPW scores on patient details $\Rightarrow$ this patient's effect; HC3 robust 95% inter-

val. **PEHE:** typical error of one patient's effect. **Coverage:** how often the 95% range contains the truth. **E-value:** how strong a hidden confounder must be to explain the effect away (1.57–2.15).

Ten likely questions

1. *Why not a normal ML model?* It predicts who does well, not what happens under a different drug; doctors choose drugs, so we need counterfactuals.
2. *Why synthetic data?* No open Indian dataset records drug and outcome; real data needs ethics approval; synthetic data has a known truth so we can measure error.
3. *Is Pima Indian data?* No — Akimel O'odham, Arizona, USA; no treatment variable. Not used.
4. *How are unsafe drugs handled?* Ten cited rules in rules.csv run first; an excluded drug is never estimated; no doses are ever shown.
5. *What if a patient is unusual?* Overlap check $\Rightarrow$ “insufficient evidence” (2.5% of test cases).
6. *Why intervals?* Every estimate is uncertain; a bare number misleads. Coverage shows ours are honest.
7. *How do you know it works?* 20 synthetic cohorts with a planted truth; bias, coverage, PEHE, regret, balance, refutation all reported.
8. *What is RAG here?* Search only licence-cleared WHO/FDA text, quote the sentences that answer, each numbered to its passage; a checker rejects any model sentence not supported by its passage; out-of-scope and dose questions are refused.
9. *Privacy?* Patient details never leave the device on the website; accounts store only name, email, role, approval.
10. *What is left?* Engine + RAG inside the chat app, embedding model + reranker (abstention 0.80 $\rightarrow$ 0.95), clinician review of the gold set and vignettes, SUS, paper.

If something fails

Wi-Fi/laptop down $\rightarrow$ play DiaCausal_demo.mp4 or open the website on a phone (works offline once opened). Unknown question $\rightarrow$ “We haven't tested that yet; this is how we would.” Never guess a number.